

Coding & Solution

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/AP24110010739/Rising-Waters
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib
Key Features Implemented	Data Collection, Data Visualization, Data Preprocessing, Decision Tree Model, Random Forest Model, KNN Model, XGBoost Model, Model Comparison, Flood Prediction, Flask Web Application
Pending / Incomplete Features	IBM Cloud Deployment, Performance Optimization, Advanced User Authentication
Setup / Run Instructions	Install required libraries using <code>pip install -r requirements.txt</code> , run <code>python app.py</code> , and open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate components for data preprocessing, model training, model evaluation, and web application development. All machine learning models are evaluated using appropriate performance metrics, and the best-performing model is integrated into the Flask application. The project structure follows standard Python coding practices, making the application maintainable, reusable, and easy to extend for future enhancements.